



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SIVASAKTHI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

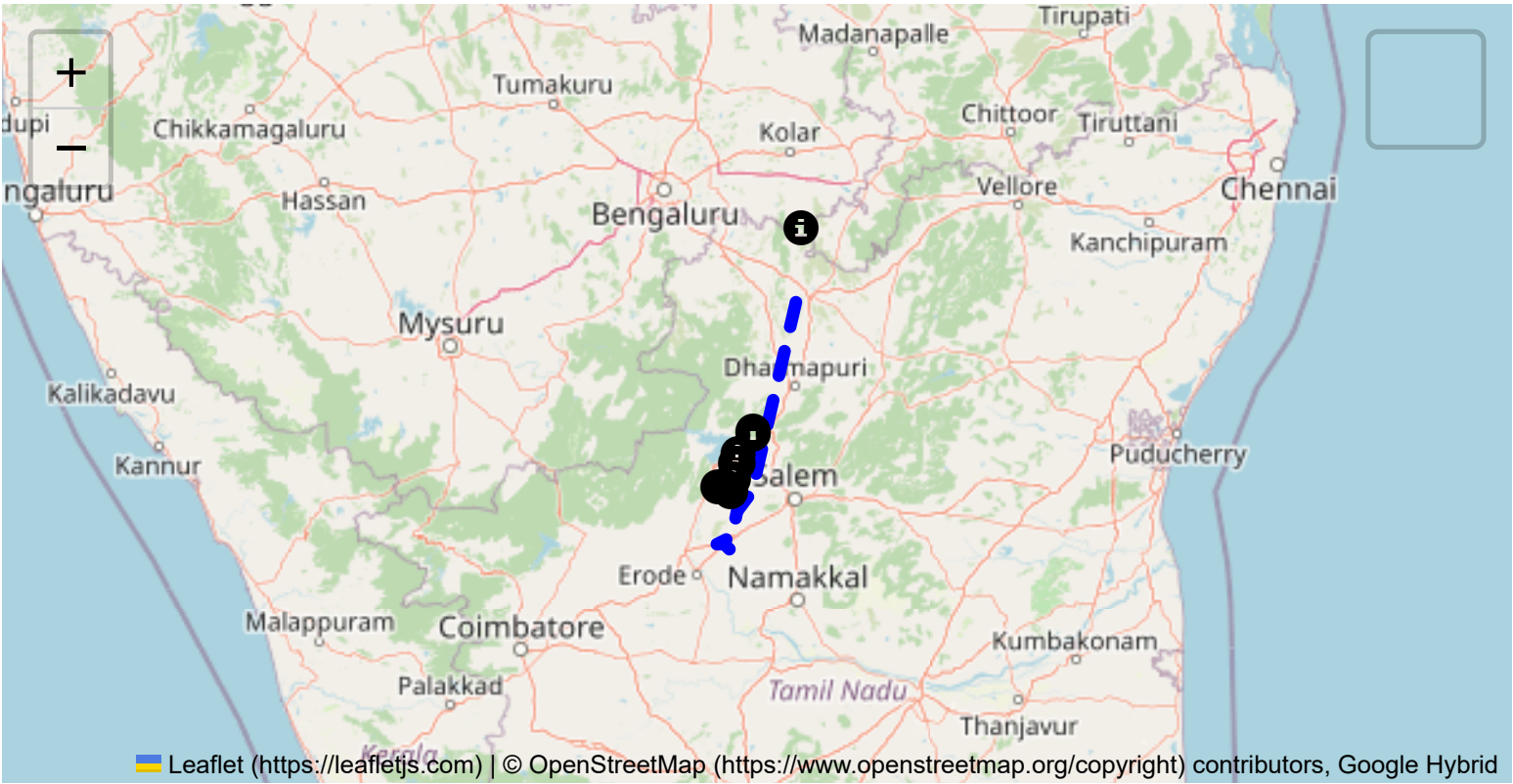
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

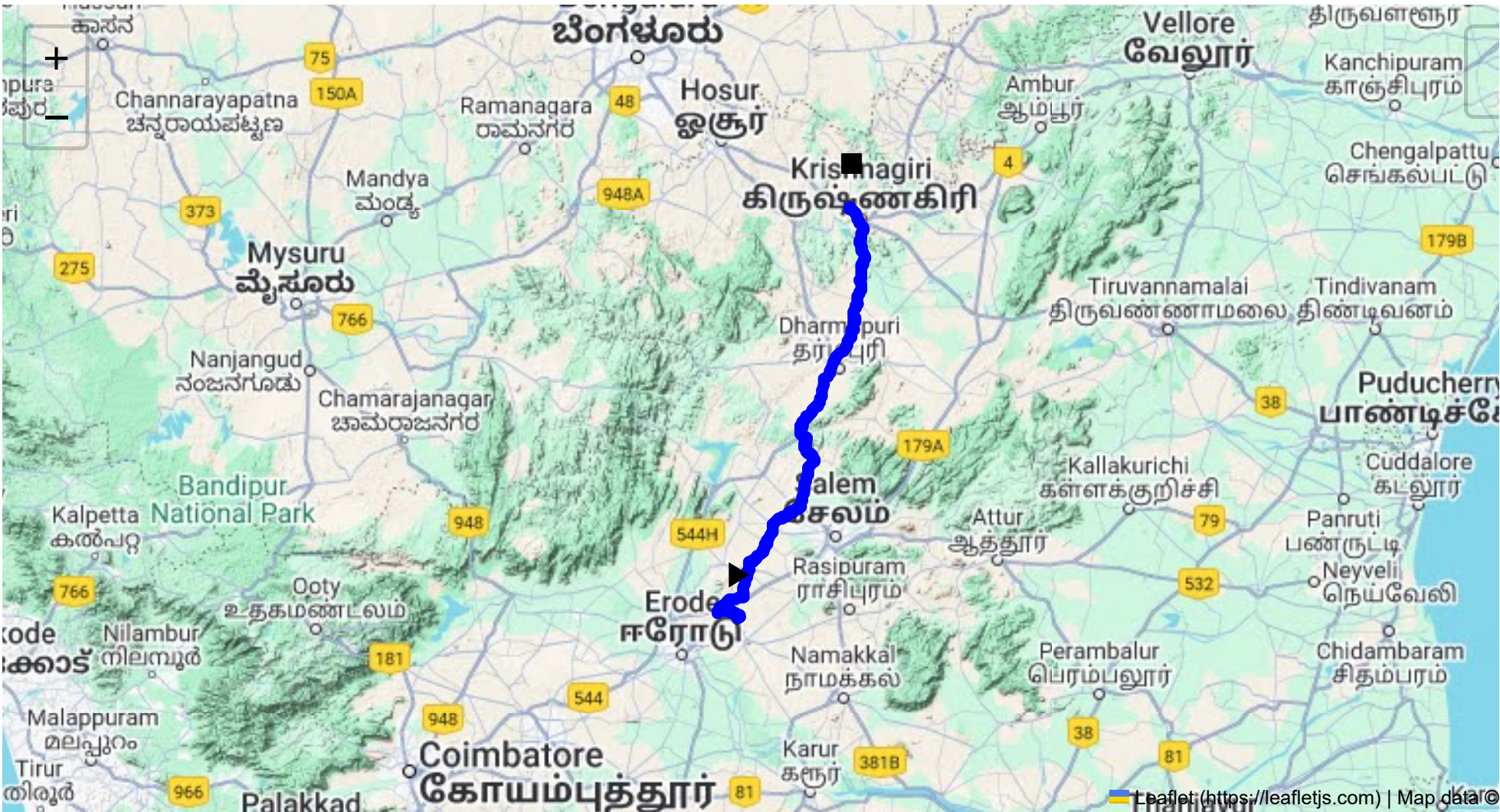
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 157.20 km
Estimated Duration: 3.0 hours
Adjusted Duration (Heavy Vehicle): 3.8 hours
Start: (11.4381, 77.8734)
End: (12.56051, 78.18277)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route begins at Sangagiri, Tamil Nadu, and traverses through a series of smaller towns and villages primarily via State Highways 86 before reaching Kundarapalli on NH 7. The route includes segments on the Sankari RS - Erode Privu Rd and Bhavani Main Road, interspersed with various towns like Padaiveedu and Erumaipatti.

2. Typical Weather Conditions and Potential Weather-Related Hazards: The region experiences a tropical climate, with monsoon seasons bringing heavy rainfall typically from June to September and lesser showers between October and November. During monsoon, road conditions might worsen due to flooding and reduced visibility, raising the risk of accidents.

3. Analysis of Traffic Patterns: Traffic can be moderate to heavy, especially in market towns and major intersections during morning (7-10 AM) and evening (4-7 PM) peak hours. Congestion-prone areas include town centers and junctions at Sankari and Tharamangalam, where local traffic is denser.

4. Assessment of Road Quality and Infrastructure: Road quality varies across the route. While NH 7 is generally well-maintained, stretches of SH 86 might have potholes and uneven surfaces, particularly post-monsoon. Certain rural sections could lack proper signage and street lighting.

5. Suggestions for Alternative Routes for Emergencies: In case of road blockages or severe congestion, alternatives include diverting along interior rural routes around congested areas like Sankari towards Edappadi, which are less trafficked but should be taken cautiously due to narrower paths and limited infrastructure.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Transport of hazardous materials may require special permits. Speed limits are lower for trucks in populated areas, and there are restrictions on transit through populated or sensitive zones during peak hours. Always carry necessary documentation for hazardous material.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials: Reports in the region highlight occasional accidents involving overturning trucks, particularly during wet conditions. However, major incidents involving hazardous materials are less documented but typically involve human error or mechanical failure.

8. Environmental Considerations and Sensitive Areas: The route passes through agricultural areas where pesticides and fertilizers are important. Transport of hazardous materials may pose a risk to these lands and nearby water bodies. Awareness and adherence to environmental regulations are critical.

9. Analysis of Communication Coverage: Major towns likely offer good mobile network coverage, while rural stretches might experience dead zones, particularly in low-lying areas or wooded sections. Truck drivers should have alternative communication means, like satellite phones, in such areas.

10. Estimated Emergency Response Times for Different Route Segments: Emergency response in urban segments (like Sankari and Tharamangalam) is relatively swift, usually within 20-30 minutes. In rural areas, response times may extend to an hour or more due to limited facilities and longer distances to emergency services.

12. Overall Summary of Risk Assessment: This 157.20-kilometer route, although well-traveled for heavy vehicles, presents moderate risks due to inconsistent road quality, weather impacts, and potential congestion. Drivers should be well-prepared with an understanding of local regulations, alternative route strategies, and effective communication plans to mitigate potential disruptions. The area's moderate emergency response capacity further underscores the importance of proactive safety measures and adherence to legal transport requirements for hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	High	11.45571, 77.81702	15 KM/Hr
7	Turn	High	11.45615, 77.81677	15 KM/Hr
8	Turn	Medium	11.46916, 77.82431	30 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.56096, 77.89931	30 KM/Hr
12	Turn	Medium	11.57649, 77.90610	30 KM/Hr
13	Turn	Medium	11.59214, 77.90656	30 KM/Hr
14	Turn	Medium	11.63833, 77.94818	30 KM/Hr
15	Turn	Medium	11.64171, 77.95184	30 KM/Hr
16	Turn	Medium	11.64243, 77.95198	30 KM/Hr
17	Turn	Medium	11.64721, 77.95118	30 KM/Hr
18	Turn	Medium	11.69559, 77.97112	30 KM/Hr
19	Turn	Medium	11.71386, 77.99552	30 KM/Hr
20	Turn	Medium	11.71735, 78.00019	30 KM/Hr
21	Turn	Medium	11.71915, 78.01106	30 KM/Hr
22	Turn	Medium	11.72492, 78.01660	30 KM/Hr
23	Turn	Medium	11.72765, 78.02291	30 KM/Hr
24	Turn	Medium	11.73764, 78.03441	30 KM/Hr
25	Turn	Medium	11.73752, 78.03679	30 KM/Hr
26	Turn	High	11.73652, 78.03866	15 KM/Hr
27	Turn	Medium	11.74257, 78.04708	30 KM/Hr
28	Turn	Medium	11.77998, 78.05786	30 KM/Hr
29	Turn	Medium	11.86915, 78.08382	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
30	Turn	Medium	11.87998, 78.08170	30 KM/Hr
31	Turn	Medium	11.91266, 78.05852	30 KM/Hr
32	Turn	Medium	11.93365, 78.06387	30 KM/Hr
33	Turn	Medium	11.93746, 78.05331	30 KM/Hr
34	Turn	Medium	11.95531, 78.04698	30 KM/Hr
35	Turn	High	11.95854, 78.04990	15 KM/Hr
36	Turn	High	11.96015, 78.04945	15 KM/Hr
37	Turn	Medium	11.96266, 78.05115	30 KM/Hr
38	Turn	High	11.96433, 78.05113	15 KM/Hr
39	Turn	High	11.96542, 78.04938	15 KM/Hr
40	Turn	High	11.96650, 78.04927	15 KM/Hr
41	Turn	Medium	12.17041, 78.16823	30 KM/Hr
42	Turn	Medium	12.22515, 78.19158	30 KM/Hr
43	Turn	Medium	12.22995, 78.19466	30 KM/Hr
44	Turn	Medium	12.24685, 78.19561	30 KM/Hr
45	Turn	Medium	12.25120, 78.20010	30 KM/Hr
46	Turn	Medium	12.25442, 78.20015	30 KM/Hr
47	Turn	Medium	12.29194, 78.19961	30 KM/Hr
48	Turn	Medium	12.32012, 78.20734	30 KM/Hr
49	Turn	Medium	12.40661, 78.22318	30 KM/Hr
50	Turn	Medium	12.42548, 78.22666	30 KM/Hr
51	Turn	Medium	12.50287, 78.22350	30 KM/Hr
52	Turn	Medium	12.50662, 78.22272	30 KM/Hr
53	Blind Spot	Blind Spot	12.56111, 78.18166	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
3	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
4	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
5	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
6	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
7	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
8	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
9	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
10	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
11	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
12	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
14	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
15	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
16	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
17	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
18	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
19	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
20	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
21	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
22	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
23	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
24	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
25	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
27	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
29	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
30	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
31	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
32	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
37	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium
38	clinic	Joe Hospital	12.5209601, 78.2150623	30 km/h	Medium
39	clinic	Ohm Prakash Clinic	12.5206321, 78.2159682	30 km/h	Medium
40	clinic	Gokulam Clinic & Scans	12.5207976, 78.2162023	30 km/h	Medium
41	clinic	Srinivasa Clinic	12.520975, 78.215755	30 km/h	Medium
42	clinic	Dr. Anbalagan Clinic	12.5208366, 78.2160593	30 km/h	Medium
43	clinic	Shri Narayana Bone & Joint Clinic	12.523617, 78.210613	30 km/h	Medium
44	clinic	AZ Family Clinic	12.5234942, 78.2127422	30 km/h	Medium
46	clinic	Shree Ramachandra ENT Centre	12.5290096, 78.2037359	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
47	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
48	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
49	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
13	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
26	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
28	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
33	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
34	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
35	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
36	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
45	school	Vivekananda School	12.5231937, 78.2121429	30 km/h	Medium
50	school	Sri Saraswathi Vidyalaya	12.5620799, 78.183049	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49395, 77.88537



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57649, 77.90610



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59214, 77.90656



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63833, 77.94818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64171, 77.95184



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64243, 77.95198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64721, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69559, 77.97112



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71386, 77.99552



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71735, 78.00019



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71915, 78.01106



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72492, 78.01660



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72765, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73764, 78.03441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73752, 78.03679



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73652, 78.03866



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74257, 78.04708



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77998, 78.05786



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86915, 78.08382



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87998, 78.08170



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91266, 78.05852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93365, 78.06387



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96266, 78.05115



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17041, 78.16823



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22515, 78.19158



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22995, 78.19466



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24685, 78.19561



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25120, 78.20010



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25442, 78.20015



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29194, 78.19961



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.32012, 78.20734



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40661, 78.22318



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.42548, 78.22666



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50287, 78.22350



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50662, 78.22272



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.56111, 78.18166

